

STAT 301 In-class Activity No. 6

Consider the “wage” dataset again. We have fitted various linear regression models in previous activities. Since “all models are wrong”, one way to confirm previous conclusions is to look at the data in a different way and consider alternative models. For example, instead of model wage directly, we may consider if an individual’s wage is above or below average. In other words, we can define a new response y for individual i as follows: $y_i = 1$ if individual i ’s wage is above average and $y_i = 0$ otherwise. Now, this new response is binary, so we cannot use a linear regression model. Instead, we should consider a logistic regression model.

Fit a logistic regression model with the new response and one predictor EDUCATION. Is EDUCATION associated with an individual’s wage being above/below average?